C. V. Raman Global University
ODISHA BHUBANESWAR INDIA

Department of Chemistry

BIOLOGY AND ENVIRONMENTAL SCIENCE (CY202)

Unit 1: Cells and Biomolecules - Structure

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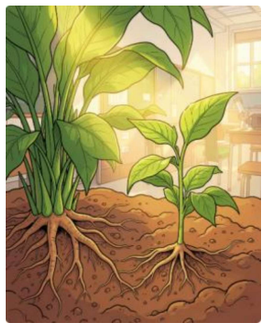
Are we ALIVE?

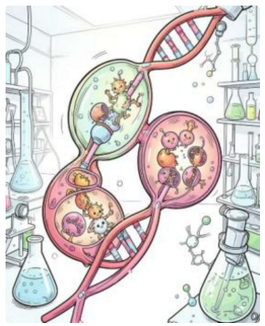
Fundamental Characteristics of Living Organisms

What separates a living organism from a non-living object? Biologists have identified a set of core properties that all living things share. Understanding these characteristics is the foundation of all biological study.



Growth

Living things increase in size by making more cells or enlarging existing ones. Plants keep growing throughout life, while animals usually stop at a set stage.



Reproduction

Life continues by making new life. Some organisms reproduce from one parent, while others combine genetic material from two parents for variation.

Metabolism

Living organisms constantly build up and break down molecules to stay alive. These chemical reactions power growth, repair, and energy use.

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More Characteristics of Life

Beyond growth, reproduction, and metabolism, living organisms share several additional defining properties that distinguish them from non-living matter.

Cellular Organization

All living organisms are composed of cells — the basic structural and functional units of life. Organization ranges from simple **unicellular** organisms (like bacteria) to complex **multicellular** organisms (like humans and trees) with specialized tissues and organs.

Consciousness

The ability to sense and respond to environmental stimuli such as light, heat, and chemicals. Plants demonstrate **phototropism** (growing toward light). Humans possess a higher level known as **self-consciousness** — awareness of one's own existence and thoughts.

Homeostasis

Living organisms maintain a stable internal environment despite changes in the external world. For example, humans regulate body temperature around 98.6°F (37°C) regardless of outside weather, through mechanisms like sweating and shivering.

Adaptation

Over generations, organisms develop structural, physiological, or behavioral modifications that improve survival in specific environments. Desert plants store water; Arctic animals grow thick fur. Adaptation is the basis of evolution by natural selection.

Movement

Movement can be **external** (locomotion — movement of limbs, walking, flying) or **internal** (cytoplasmic streaming inside cells, transport of molecules). Even plants that appear stationary exhibit internal movement at the cellular level.

Life Span

Every organism has a definite life span — a period from birth (or germination) to natural death. Life spans vary enormously: mayflies live 24 hours, while some trees live thousands of years. Death is a natural and inevitable part of the life cycle.

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Organic Constituents of Cells

Cells are the fundamental units of life, and their function depends on a sophisticated array of organic macromolecules. This presentation explores the three major classes of organic cellular constituents: **carbohydrates**, **lipids**, and **proteins.** These biomolecules are central to energy metabolism, structural integrity, enzymatic catalysis, and cellular signaling. A rigorous understanding of their chemistry, classification, and biological roles is essential for any graduate student in the life sciences.

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CARBOHYDRATES — PART I

Introduction to Carbohydrates & Monosaccharides

Carbohydrates are macronutrients composed of **carbon**, **hydrogen**, and **oxygen**, typically in a 1:2:1 ratio (empirical formula: $C_n(H_2O)_n$). They are the primary energy currency of living systems and are classified by the degree of polymerization into monosaccharides, disaccharides, oligosaccharides, and polysaccharides. During digestion, complex carbohydrates are hydrolyzed into **glucose** (blood sugar), the principal fuel for cellular respiration.

Monosaccharides: The Fundamental Units

Monosaccharides are the simplest carbohydrates — single sugar units that cannot be hydrolyzed further. They are classified by carbon count: **trioses** (3C), **tetroses** (4C), **pentoses** (5C), and **hexoses** (6C). Key hexoses include **glucose** (primary metabolic fuel), **fructose** (ketohexose in fruits), and **galactose** (component of lactose). In aqueous solution, monosaccharides exist in equilibrium between open-chain and cyclic (pyranose/furanose) forms, a property critical to their reactivity.

Structural Organization

Monosaccharides contain multiple hydroxyl (–OH) groups and either an aldehyde (aldoses) or ketone (ketoses) functional group. Glucose, for instance, is an aldohexose that cyclizes to form glucopyranose. The stereochemistry at each carbon center gives rise to numerous isomers with distinct biochemical properties. **Mannose**, for example, differs from glucose only at C-2, yet plays unique roles in glycoprotein biosynthesis and cell–cell recognition.

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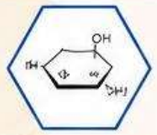
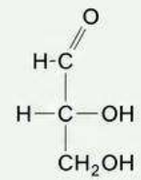
MONOSACCHARIDES: The Simplest Carbohydrates



Monosaccharides are the simplest carbohydrates — **single sugar units that cannot be hydrolyzed further.**

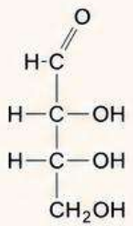
CLASSIFIED BY CARBON COUNT

Trioses (3C)



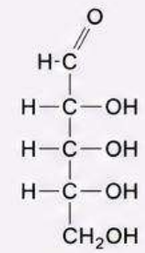
Glyceraldehyde (an example)

Tetroses (4C)



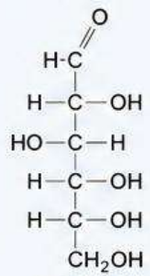
Erythrose (an example)

Pentoses (5C)



Ribose (an example)

Hexoses (6C)

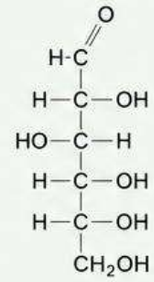


Glucose (an example)

KEY HEXOSES

Glucose (aldohexose)

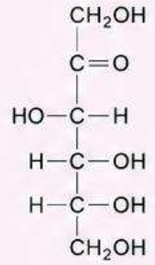
Primary metabolic fuel



D-Glucose

Fructose (ketohexose)

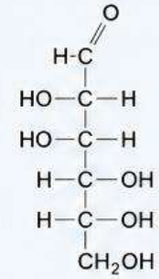
Ketohexose in fruits



D-Fructose

Galactose (aldohexose)

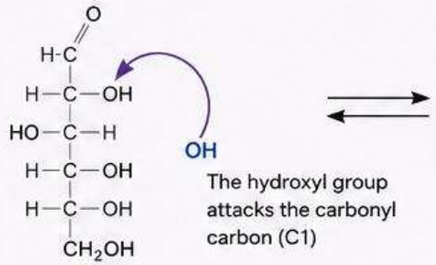
Component of lactose (milk sugar)



D-Galactose

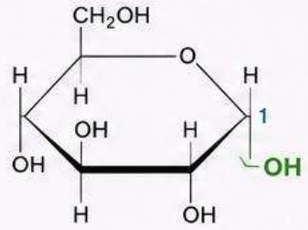
IN AQUEOUS SOLUTION, MONOSACCHARIDES EXIST IN EQUILIBRIUM BETWEEN OPEN-CHAIN AND CYCLIC FORMS

1 Open-chain form (linear)



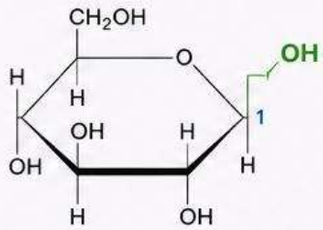
D-Glucose (open chain)

2a Cyclic (pyranose) form 6-membered ring



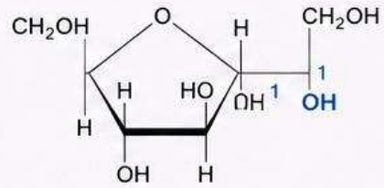
α-D-Glucopyranose

2b Cyclic (pyranose) form 6-membered ring



β-D-Glucopyranose

2c Cyclic (furanose) form 5-membered ring



α-D-Glucofuranose

These interconverting forms (mutarotation) are in dynamic equilibrium in water.
This property is **critical to the reactivity, recognition, and function of monosaccharides.**

Pyranose = 6-membered ring (like pyran)
Furanose = 5-membered ring (like furan)

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CARBOHYDRATES — PART II

Disaccharides, Oligosaccharides & Polysaccharides

Beyond monosaccharides, carbohydrates form increasingly complex structures through **glycosidic linkages.** The nature of these bonds — including their position (α or β) and the carbons involved — determines the physical and metabolic properties of the resulting molecule.

Disaccharides

Two monosaccharides linked by a glycosidic bond. **Sucrose** (glucose + fructose, α-1,2) is common table sugar. **Lactose** (glucose + galactose, β-1,4) is the primary sugar in milk. **Maltose** (glucose + glucose, α-1,4) is a product of starch digestion. Hydrolysis by specific disaccharidases is required for absorption.

Oligosaccharides

Composed of 2–9 monosaccharide units. Oligosaccharides are critical in **cell recognition** and **signal transduction.** Glycoproteins and glycolipids on cell surfaces display oligosaccharide chains that mediate immune recognition, cell adhesion, and receptor binding. ABO blood group antigens are classic examples of oligosaccharide-mediated specificity.

Polysaccharides

Long chains of monosaccharides. **Starch** (amylose + amylopectin) is the plant energy reserve; **glycogen** is the animal equivalent, stored in liver and muscle. **Cellulose** (β-1,4-linked glucose) provides structural support in plant cell walls. **Chitin** forms fungal cell walls and arthropod exoskeletons.

Clinical Note: Lactose intolerance results from deficient lactase enzyme activity, leading to undigested lactose fermentation by gut microbiota — a condition affecting ~65% of the global adult population.

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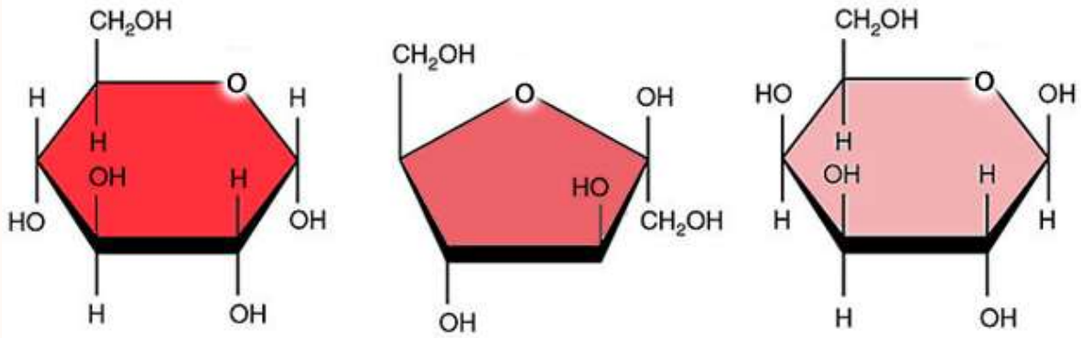
Monosaccharides



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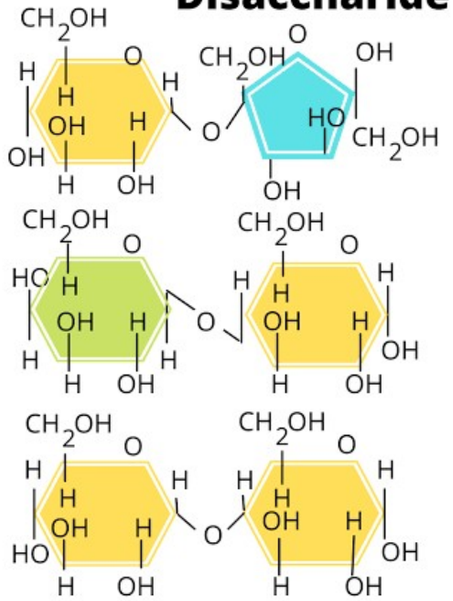
Disaccharide Examples



Sucrose
glucose + fructose

Lactose
galactose + glucose

Maltose
glucose + glucose

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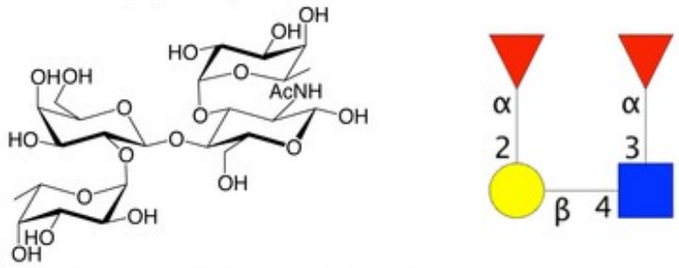
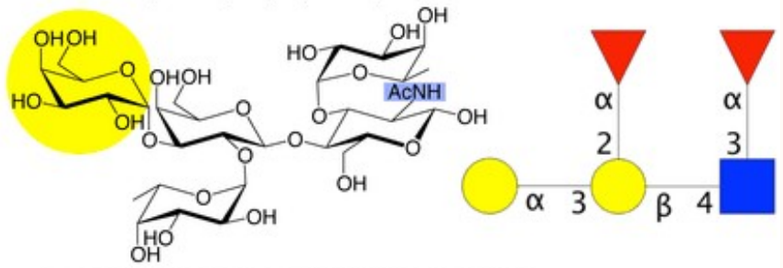
H-tetrasaccharide-BGA (H-tetra-BGA)

Fucα2Galβ4(Fucα3)GlcNAc



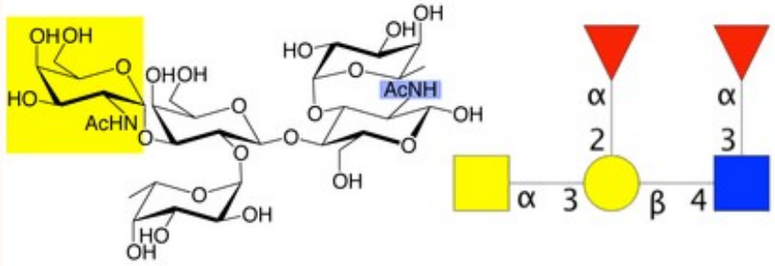
B-pentasaccharide-BGA (B-penta-BGA)

Galα3(Fucα2)Galβ4(Fucα3)GlcNAc



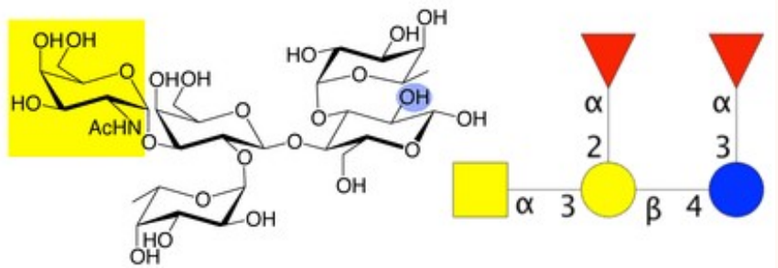
A-pentasaccharide-BGA (A-penta-BGA)

GalNAcα3(Fucα2)Galβ4(Fucα3)GlcNAc



A-pentasaccharide-HMO (A-penta-HMO)

GalNAcα3(Fucα2)Galβ4(Fucα3)Glc



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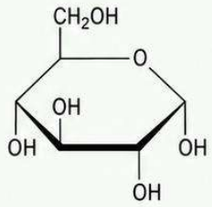
Carbohydrates: From Simple to Complex

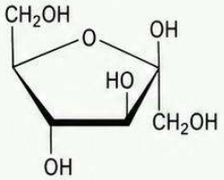
1. MONOSACCHARIDES

Monosaccharides are **single sugar units** that cannot be hydrolyzed further.

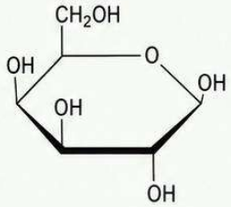
α-D-Glucopyranose



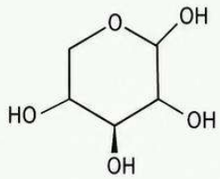
β-D-Fructofuranose



β-D-Galactopyranose



D-Xylopyranose

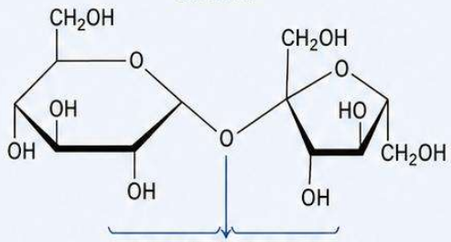


Simple to Complex

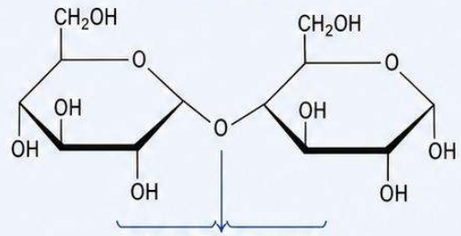
2. DISACCHARIDES

Disaccharides consist of two monosaccharides joined by a glycosidic bond.

Sucrose



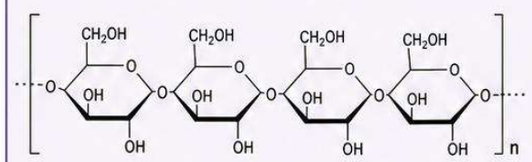
α-Maltose



3. POLYSACCHARIDES

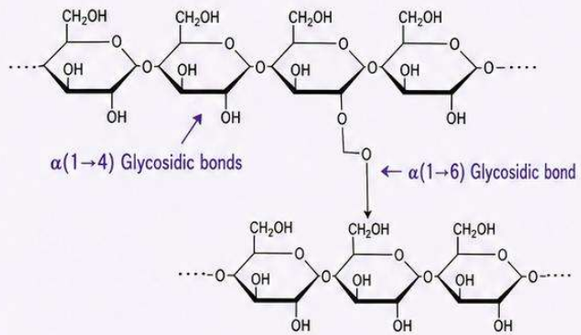
Polysaccharides are long chains of monosaccharides (hundreds to thousands of units).

Amylose



Linear polymer of α-D-glucose
α(1→4) Glycosidic bonds

Amylopectin



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CARBOHYDRATES — PART III

Functions & Dietary Sources of Carbohydrates

Key Biological Functions

- **Primary Energy Source:** Glucose oxidation via glycolysis and the citric acid cycle yields ATP, the universal energy currency. The brain alone consumes ~120 g of glucose per day.
- **Sparing Effect:** Adequate carbohydrate intake prevents gluconeogenesis from amino acids, preserving muscle protein, and inhibits ketosis by ensuring complete fatty acid oxidation.
- **Enzymatic Digestion:** Salivary and pancreatic **amylase** hydrolyze starch into maltose and dextrins; brush-border enzymes complete digestion to absorbable monosaccharides.
- **Structural Roles:** Glycosaminoglycans (e.g., hyaluronic acid, chondroitin sulfate) form the extracellular matrix, providing mechanical support and hydration.
- **Cell Signaling:** Glycoproteins and glycolipids on cell membranes mediate intercellular communication, immune recognition, and pathogen binding.

Dietary Sources by Carbohydrate Type

Fructose

Found in fruits, honey, and high-fructose corn syrup. Absorbed via GLUT5 transporters in the small intestine.

Galactose

Present in dairy products as part of lactose. Converted to glucose in the liver via the Leloir pathway.

Lactose

Abundant in milk and dairy products. Requires lactase for hydrolysis; deficiency is the most common enzyme deficiency worldwide.

Maltose & Sucrose

Maltose in cereals, beer, and potatoes; sucrose from sugar cane, sugar beets, and honey. Both require specific disaccharidases for digestion.

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LIPIDS — PART I

Lipids: Properties & Classification

Lipids are a **heterogeneous family of organic compounds** defined by their solubility in nonpolar solvents (e.g., chloroform, ether) and insolubility in water. Composed primarily of hydrocarbon chains, lipids serve as energy reserves, structural membrane components, and signaling molecules. They yield approximately **9 kcal/g** upon oxidation — more than twice the energy density of carbohydrates or proteins.

Physical Properties

Lipids are nonpolar, hydrophobic molecules stored in **adipose tissue.** Their amphipathic nature (hydrophobic tail + hydrophilic head in phospholipids) enables spontaneous formation of bilayers — the structural basis of all biological membranes.

Energy-Rich Molecules

Triacylglycerols (triglycerides) are the primary storage form of lipids. Each molecule consists of a glycerol backbone esterified to three fatty acid chains. Saturated fatty acids (no double bonds) are solid at room temperature; unsaturated fatty acids (one or more double bonds) are liquid (oils).

Membrane Architecture

Phospholipids, glycolipids, and cholesterol form the **lipid bilayer** that compartmentalizes the cell from its environment. Cholesterol modulates membrane fluidity, while glycolipids participate in cell recognition. Membrane proteins embedded in this bilayer mediate transport, signaling, and enzymatic activity.

Signaling & Regulation

Steroid hormones (e.g., cortisol, estrogen, testosterone) are derived from cholesterol. Eicosanoids (prostaglandins, leukotrienes) are signaling molecules derived from arachidonic acid that regulate inflammation, blood clotting, and immune responses.

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LIPIDS — PART II

Structure of Lipids: Fatty Acids & Complex Lipids

Fatty Acid Structure

Fatty acids are the monomeric building blocks of most lipids. They consist of a long, nonpolar **hydrocarbon chain** (typically 4–28 carbons) terminating in a polar **carboxyl group** (–COOH). Fatty acids are classified as:

- **Saturated:** No carbon–carbon double bonds (e.g., palmitic acid C16:0, stearic acid C18:0). Pack tightly, solid at room temperature.
- **Monounsaturated:** One double bond (e.g., oleic acid C18:1, omega-9). Liquid at room temperature.
- **Polyunsaturated:** Multiple double bonds (e.g., linoleic acid C18:2, omega-6; α-linolenic acid C18:3, omega-3). Essential fatty acids must be obtained from the diet.

The position and geometry (cis vs. trans) of double bonds profoundly affect membrane fluidity and metabolic function. **Trans fats**, formed during partial hydrogenation, are associated with cardiovascular disease.

Complex Lipid Classes

Triacylglycerols

Three fatty acids esterified to glycerol. Primary form of energy storage in adipocytes. Hydrolyzed by lipases during fasting or exercise.

Phospholipids

Glycerol + two fatty acids + phosphate group + polar head group (e.g., choline in phosphatidylcholine). Form the lipid bilayer of cell membranes.

Steroids

Four-ring structure (three six-membered, one five-membered). Cholesterol is the precursor of all steroid hormones, bile acids, and vitamin D.

Lipoproteins

Complexes of lipids and proteins (apolipoproteins) that transport hydrophobic lipids in aqueous blood. Classes include HDL, LDL, VLDL, and chylomicrons.

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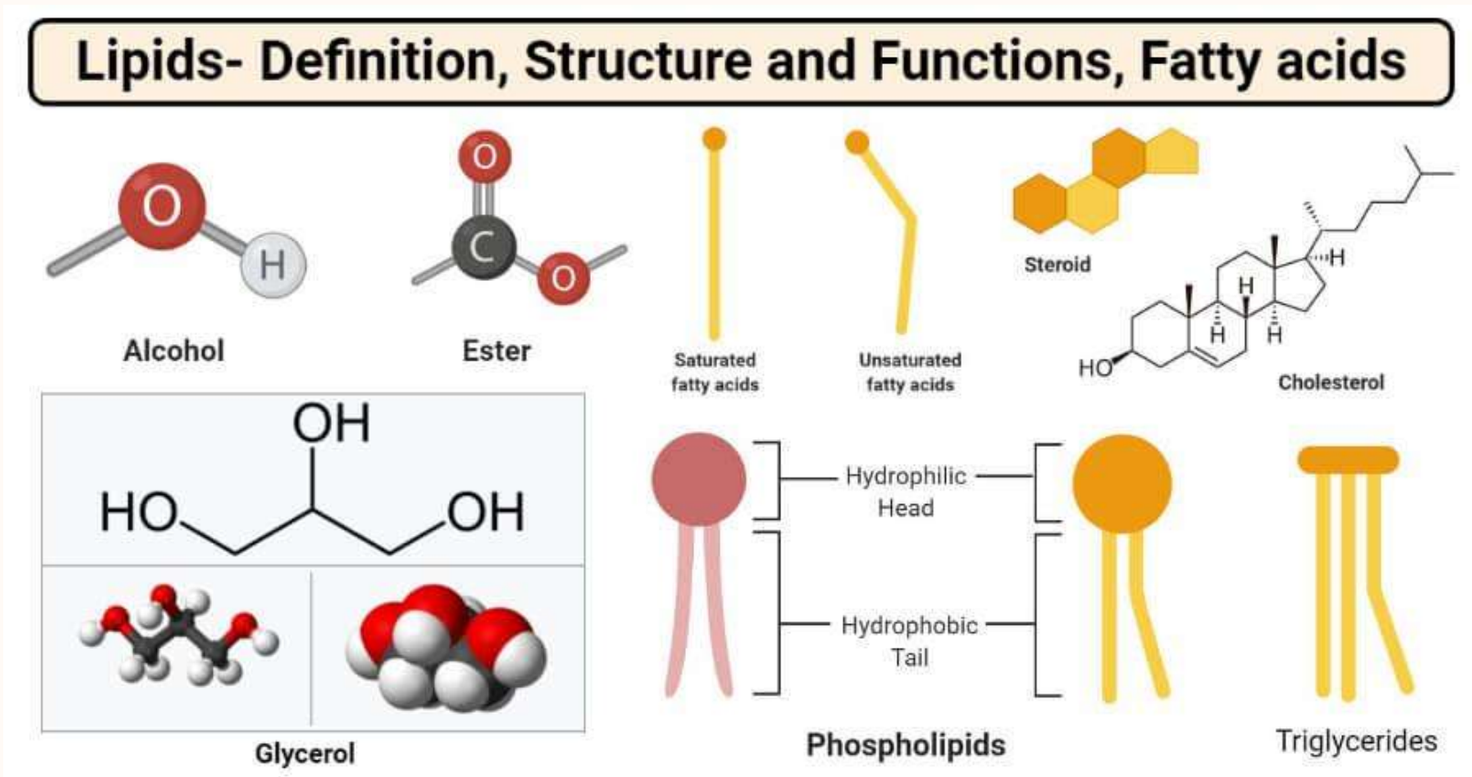
TYPES OF LIPIDS

STEROIDS

- Steroids are lipids that do not contain fatty acids.
- Examples of steroids are cholesterol, testosterone, estrogen and progesterone.

WAXES

Wax contains one molecule of alcohol that combines with another molecule of fatty acid and is waterproof.



FATS

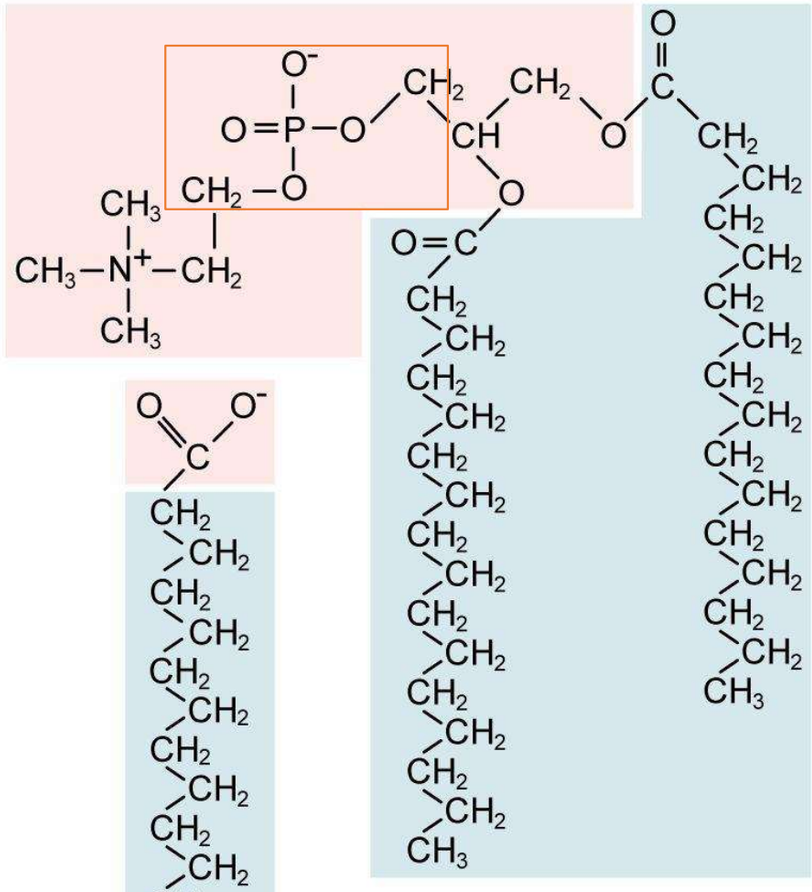
- Are triglycerides which formed from the condensation of one glycerol molecule with three molecules of fatty acids.
- There are two types of fatty acids; saturated fatty acids and unsaturated fatty acids.

PHOSPHOLIPIDS

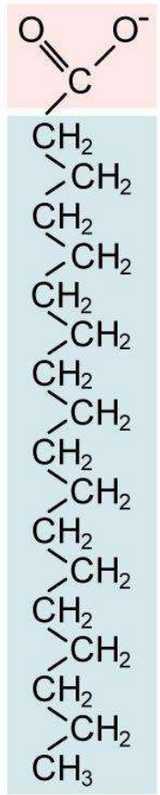
Phospholipids are a major component of plasma membranes and are made up of one molecule of glycerol that combines with two molecules of fatty acid and one group of phosphate.

structural formula

phospholipid (phosphatidylcholine)

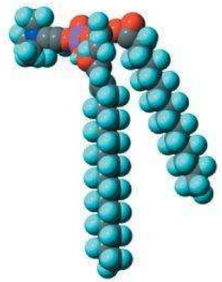


fatty acid (stearic acid)



space-filling model

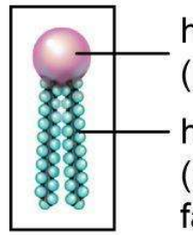
phospholipid molecule



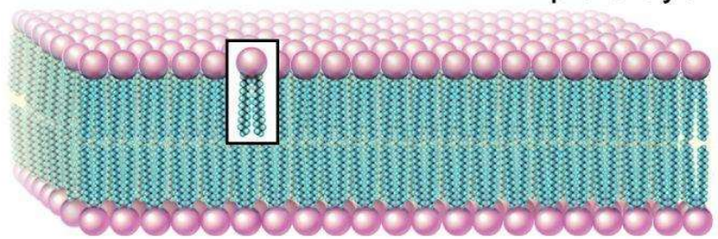
fatty acid molecule

generic simplified depiction

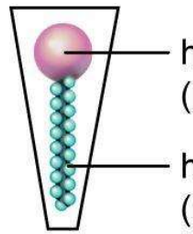
phospholipid molecule

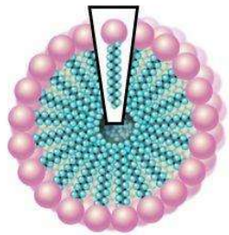


lipid bilayer



fatty acid molecule





micelle

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PROTEINS — PART I

Introduction to Proteins & Amino Acids

Proteins are the most functionally diverse class of biomolecules, serving as **enzymes**, **structural scaffolds**, **transporters**, **receptors**, **antibodies**, and **signaling molecules**. They are polymers of **amino acids** linked by **peptide bonds** (amide linkages formed by condensation between the α-carboxyl and α-amino groups). The human proteome comprises an estimated 20,000–25,000 distinct proteins, each with a unique three-dimensional structure determined by its amino acid sequence.

Amino Acid Structure

All 20 standard amino acids share a common backbone: a central **α-carbon** bonded to an amino group ($–NH_2$), a carboxyl group (–COOH), a hydrogen atom, and a variable **R group** (side chain). The chemical nature of the R group — nonpolar, polar uncharged, acidic, or basic — determines the amino acid's properties and its role in protein structure and function.

Peptide Bond Formation

Peptide bonds are formed by **dehydration synthesis**, releasing water. The resulting polypeptide chain has directionality: an **N-terminus** (free amino group) and a **C-terminus** (free carboxyl group). Protein synthesis proceeds from N- to C-terminus on ribosomes, guided by mRNA templates during translation.

Essential vs. Non-Essential

Of the 20 standard amino acids, **9 are essential** (histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine) and must be obtained from the diet. The remaining 11 are synthesized endogenously. **Conditionally essential** amino acids (e.g., arginine, glutamine) become required during illness or rapid growth.

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TWENTY-ONE PROTEINOGENIC α-AMINO ACIDS

Side chain charge at physiological pH 7.4

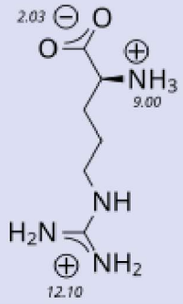
pK_a values shown italicized

⊕ Positive
⊖ Negative

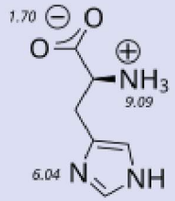
A. Amino Acids with Electrically Charged Side Chains

Positive

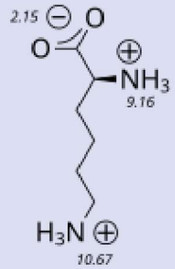
Arginine Arg R



Histidine His H

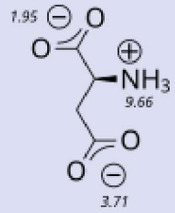


Lysine Lys K

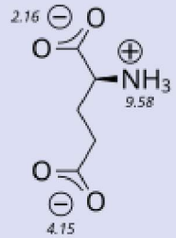


Negative

Aspartic Acid Asp D

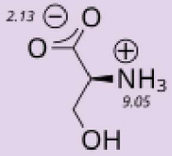


Glutamic Acid Glu E

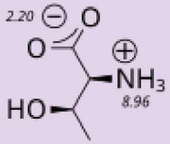


B. Amino Acids with Polar Uncharged Side Chains

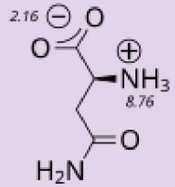
Serine Ser S



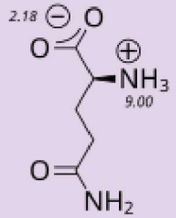
Threonine Thr T



Asparagine Asn N

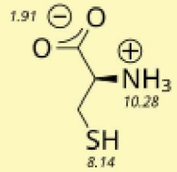


Glutamine Gln Q

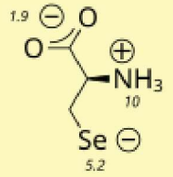


C. Special Cases

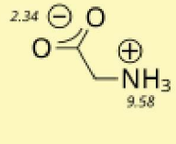
Cysteine Cys C



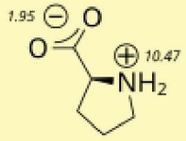
Selenocysteine Sec U



Glycine Gly G

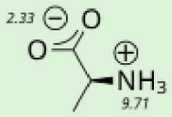


Proline Pro P

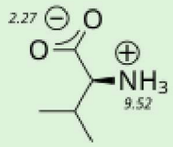


D. Amino Acids with Hydrophobic Side Chains

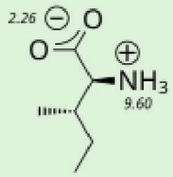
Alanine Ala A



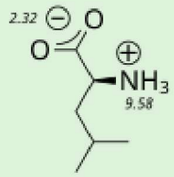
Valine Val V



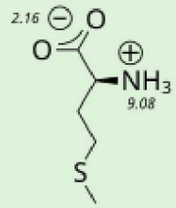
Isoleucine Ile I



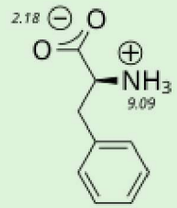
Leucine Leu L



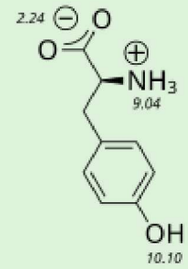
Methionine Met M



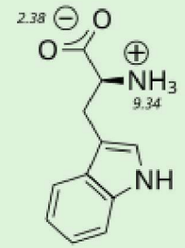
Phenylalanine Phe F



Tyrosine Tyr Y



Tryptophan Trp W



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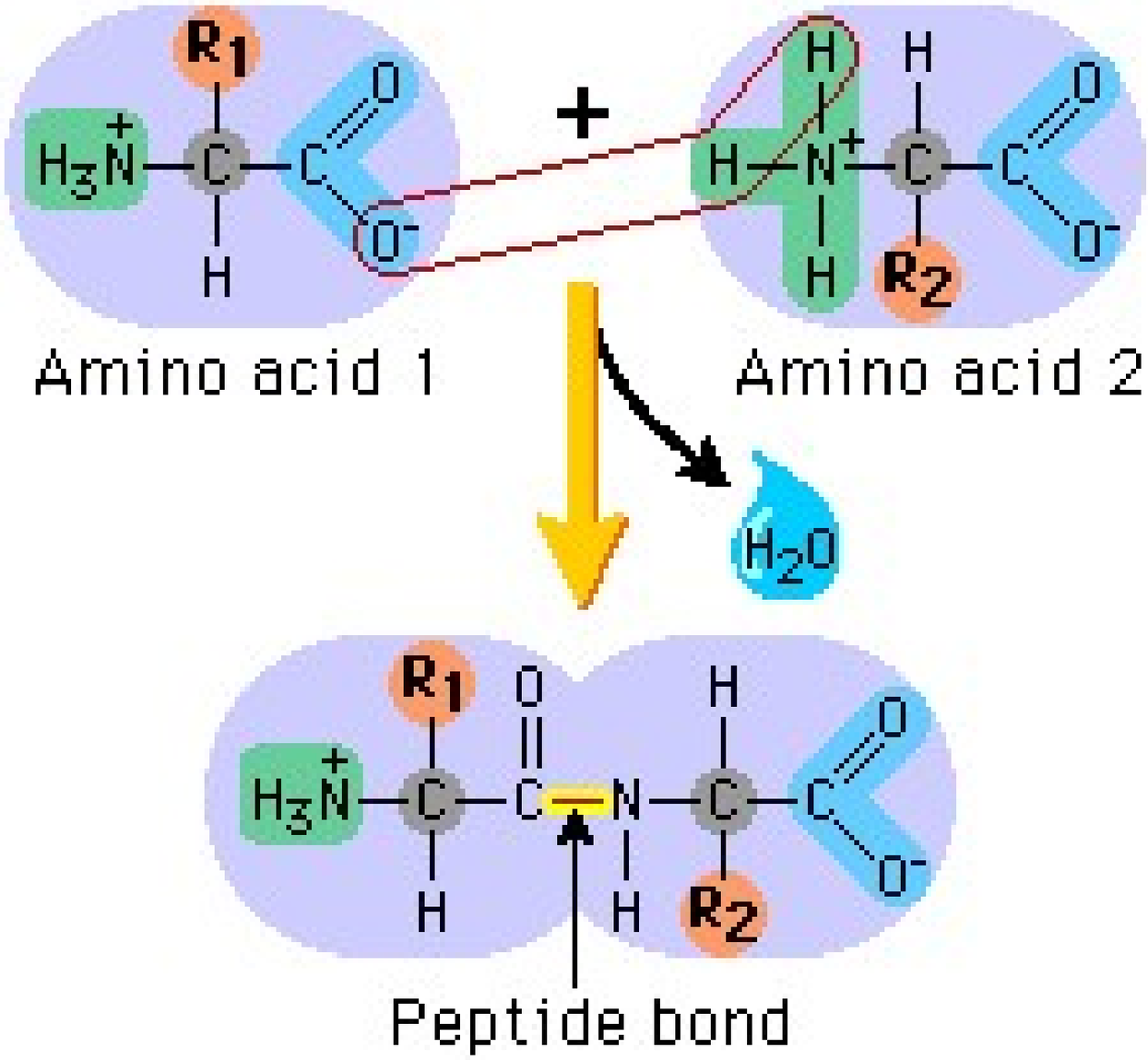

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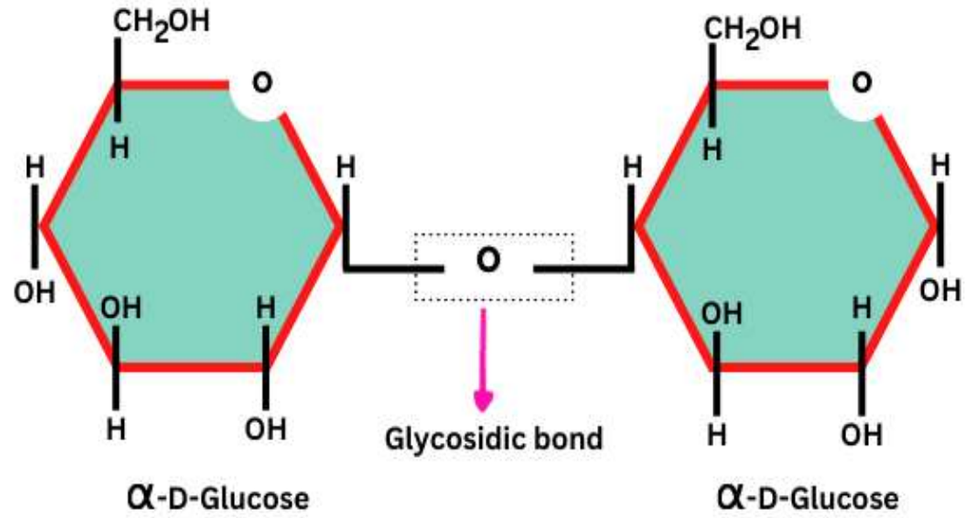

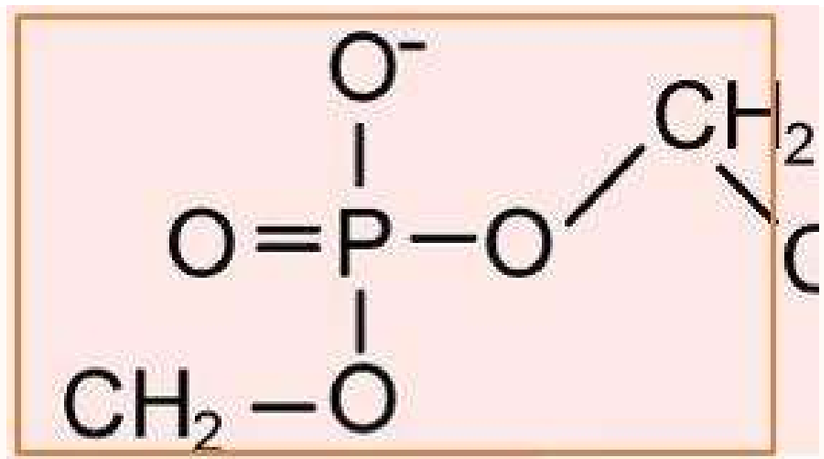

PHOSPHODIESTER BOND

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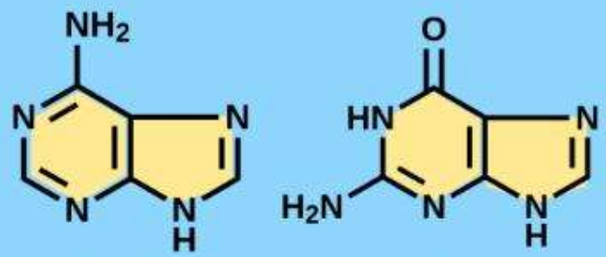
Basic DNA STRUCTURE

Purines and Pyrimidines

PURINES

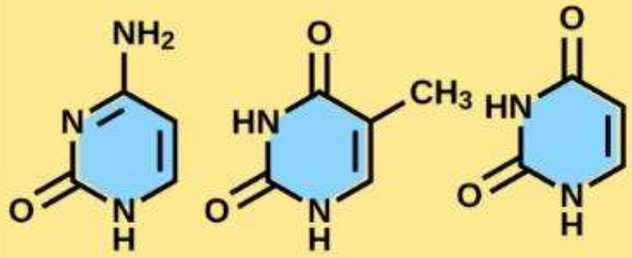


adenine guanine

6 + 5 membered rings

Heterocyclic aromatic compound with a pyrimidine ring fused with an imidazole ring

PYRIMIDINES

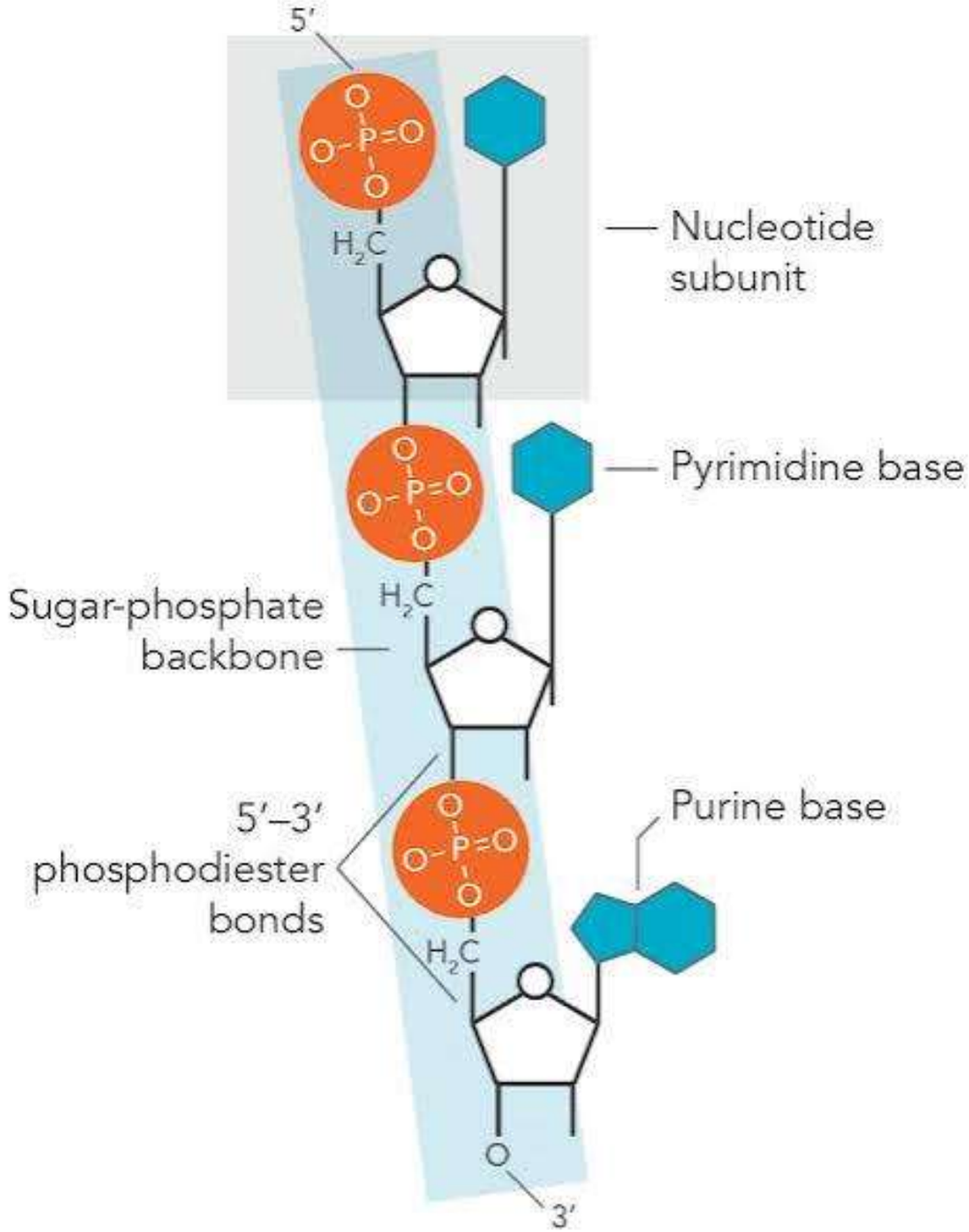


cytosine thymine uracil

6 membered ring

Heterocyclic aromatic compound with a pyrimidine ring, which has nitrogen atoms at the 1 and 3 positions

sciencenotes.org



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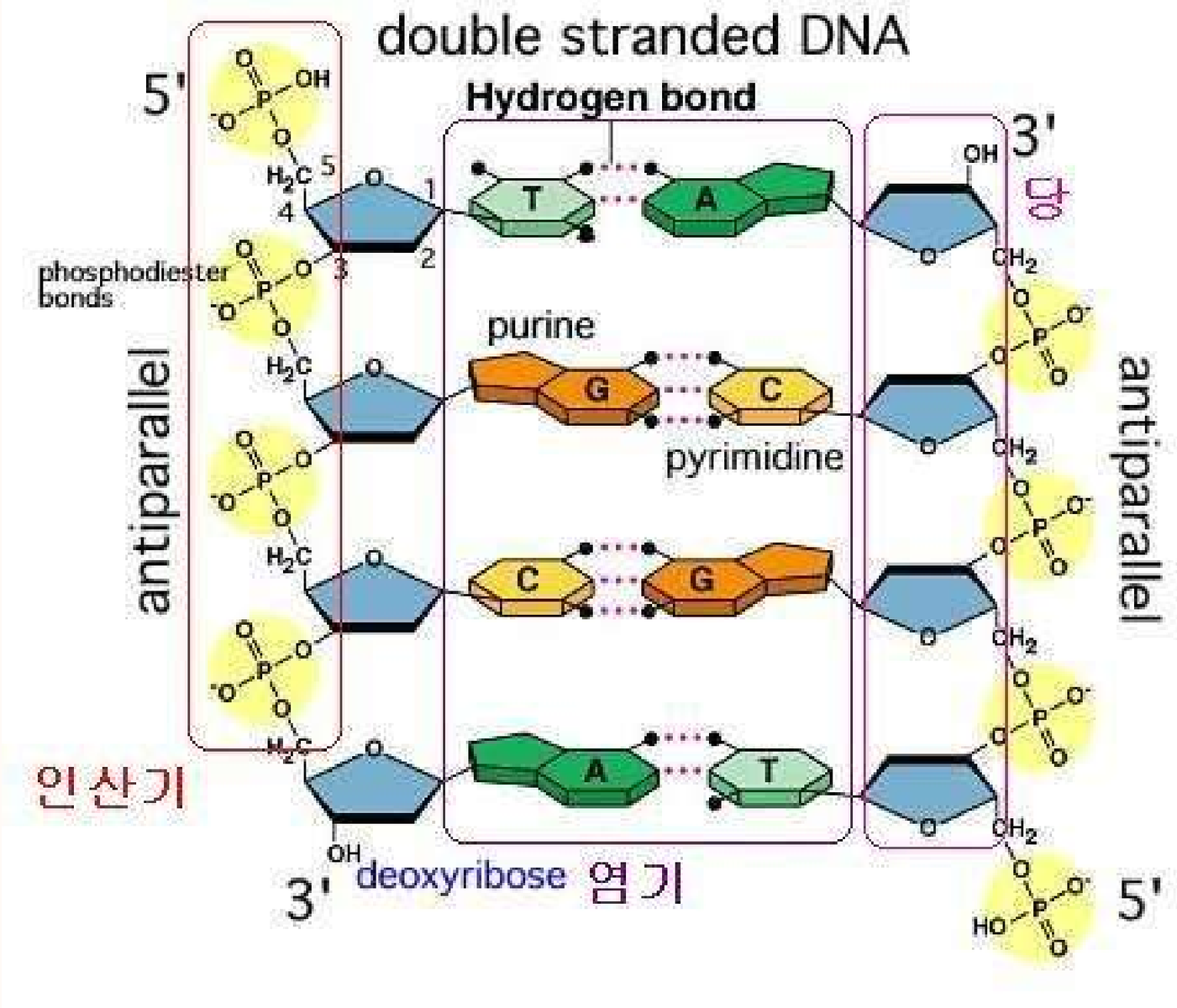
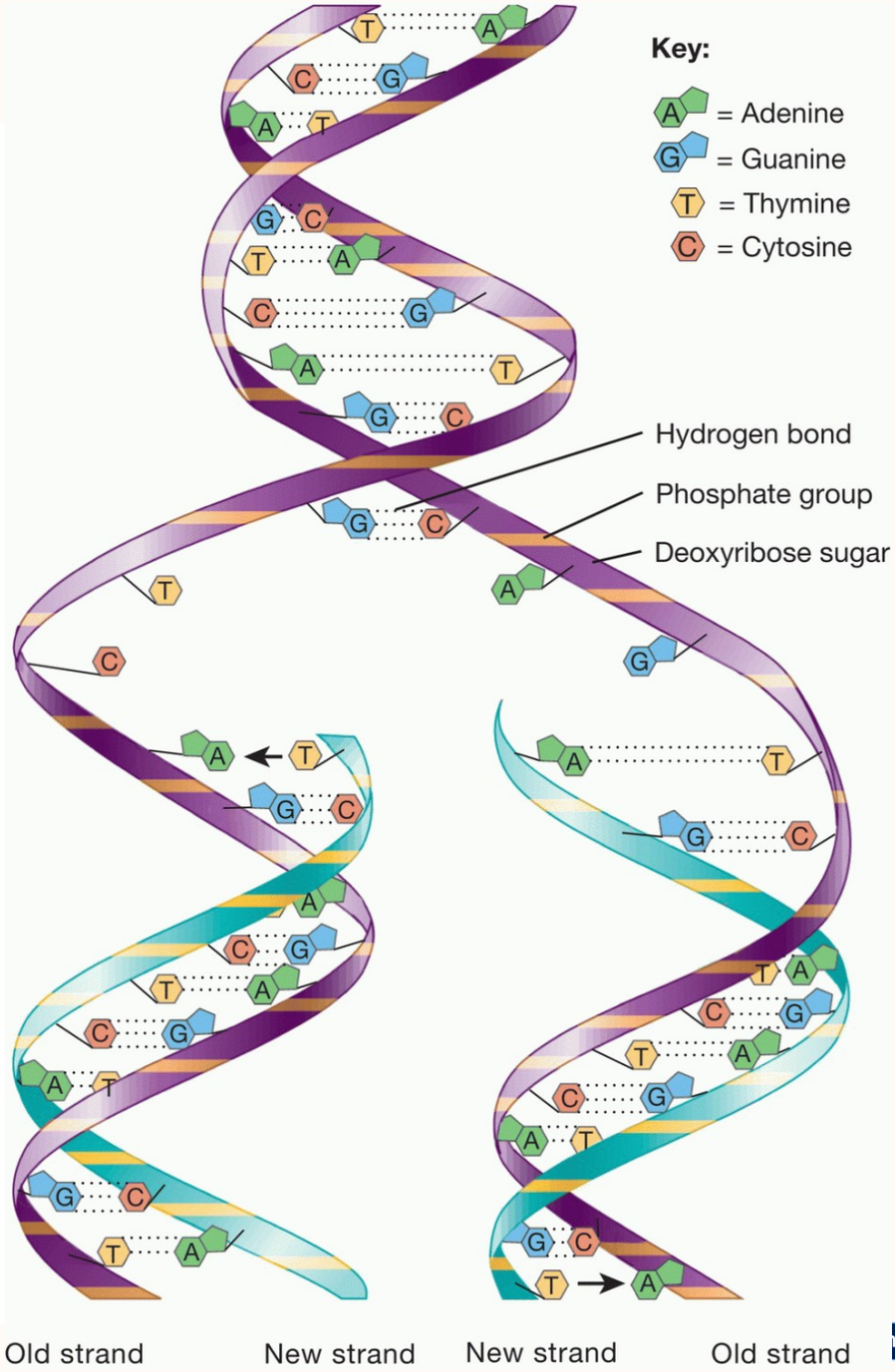

double stranded DNA
Hydrogen bond
5'
3'
phosphodiester bonds
antiparallel
antiparallel
purine
pyrimidine
인산기
deoxyribose 염기
3'
5'
DNA


Key:
= Adenine
= Guanine
= Thymine
= Cytosine
Hydrogen bond
Phosphate group
Deoxyribose sugar
Old strand
New strand
New strand
Old strand

PROTEINS — PART II

The Four Levels of Protein Structure

Protein function is inseparable from structure. The hierarchical organization of proteins into four structural levels — primary, secondary, tertiary, and quaternary — determines their three-dimensional conformation and, consequently, their biological activity. Disruption of this structure (**denaturation**) abolishes function.

1

Quaternary Structure

Association of multiple polypeptide subunits (e.g., hemoglobin = 2α + 2β chains; DNA polymerase). Stabilized by non-covalent interactions and sometimes disulfide bridges between subunits.

2

Tertiary Structure

Three-dimensional folding of the entire polypeptide chain. Stabilized by hydrophobic interactions, hydrogen bonds, ionic bonds, disulfide bridges, and van der Waals forces. Creates active sites and binding pockets.

3

Secondary Structure

Local folding patterns: **α-helices** (right-handed coils stabilized by backbone H-bonds every 4 residues) and **β-sheets** (extended strands connected by H-bonds). Also includes turns and random coils.

4

Primary Structure

The linear sequence of amino acids in the polypeptide chain, encoded by the gene. The primary structure contains all information necessary to specify the final three-dimensional conformation (Anfinsen's dogma).

⚠ **Denaturation:** Heat, extreme pH, organic solvents, or detergents can disrupt non-covalent interactions, causing proteins to unfold and lose function. Some proteins (e.g., ribonuclease A) can refold spontaneously; others require molecular chaperones (e.g., HSP70, chaperonins).

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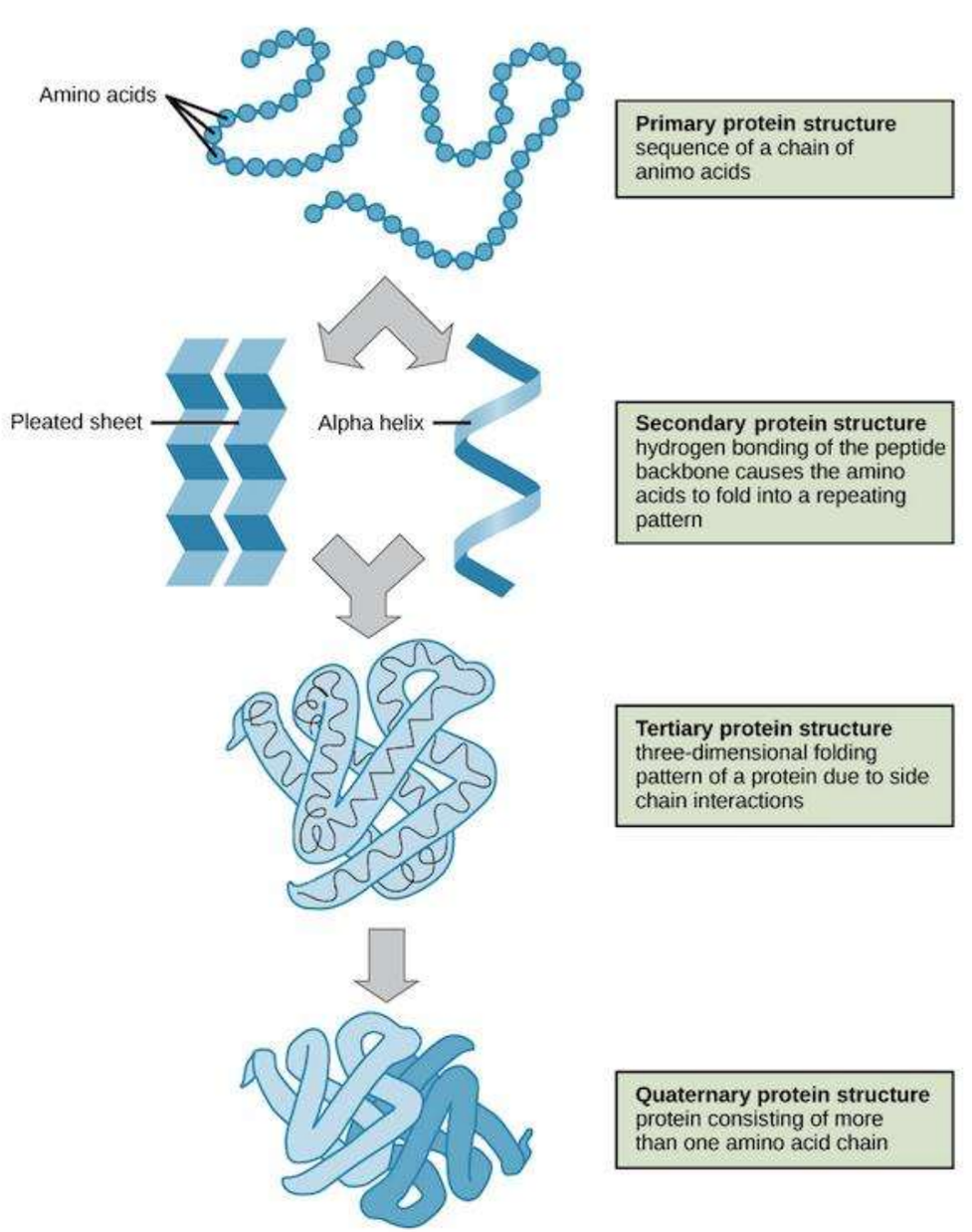
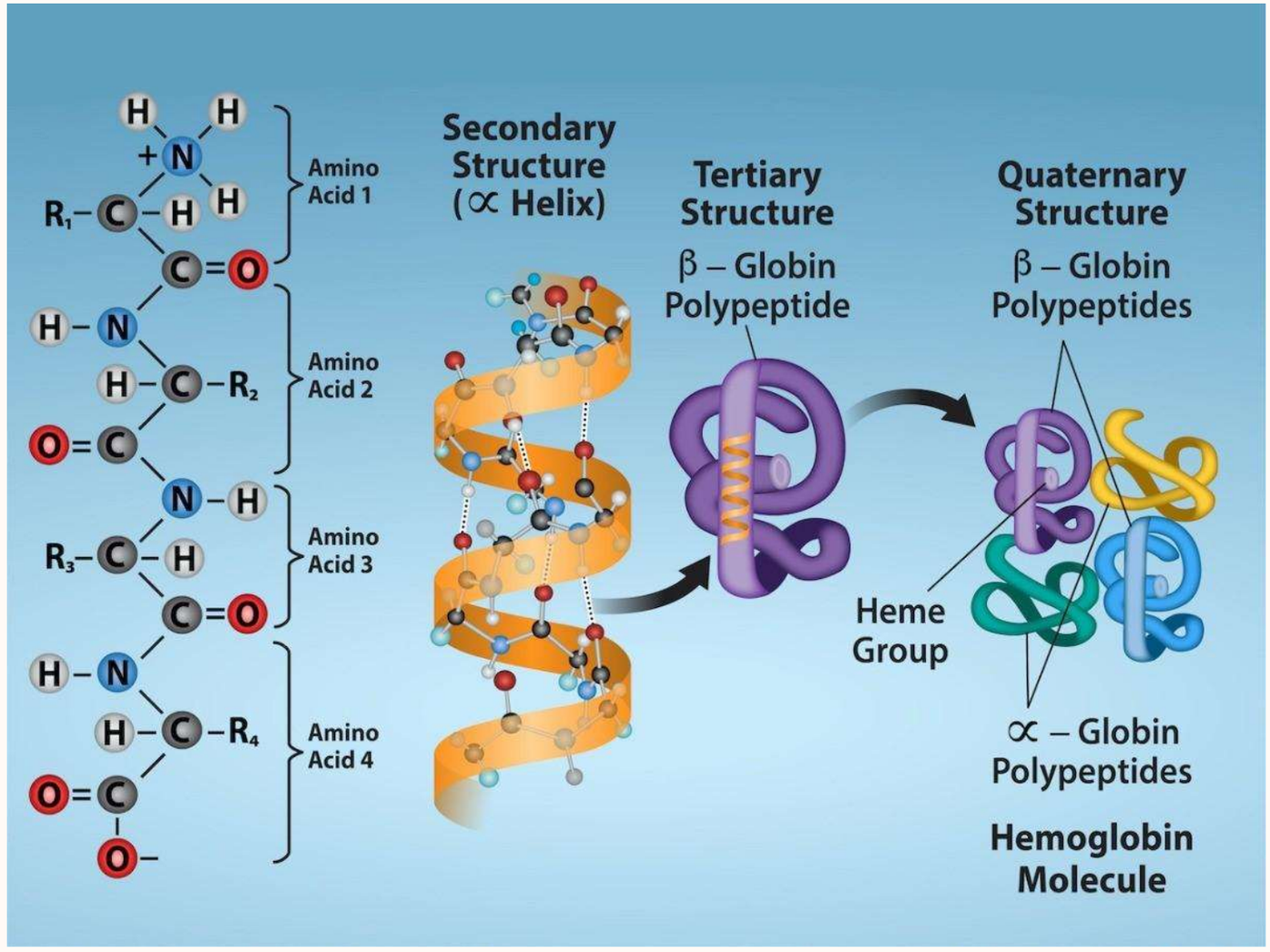
The Four Levels of Protein Structure





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CLASSIFICATION OF PROTEINS

Based on Composition, Shape, and Function

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1. BASED ON COMPOSITION

SIMPLE PROTEINS

Yield only amino acids on hydrolysis

- **Examples:** Albumin, Globulin, Histones

CONJUGATED PROTEINS

Contain a non-protein component (prosthetic group)

- **Examples:**
  - **Glycoprotein** (protein + carbohydrate)
  - **Lipoprotein** (protein + lipid)
  - **Metalloprotein** (protein + metal ion)

DERIVED PROTEINS

Formed by partial hydrolysis or denaturation of simple or conjugated proteins

- **Examples:** Peptones, Proteoses

2. BASED ON SHAPE

FIBROUS PROTEINS

Long, thread-like shape

- Insoluble in water
- **Function:** Structural support
- **Examples:** Collagen (skin), Keratin (hair)

GLOBULAR PROTEINS

Spherical or globular shape

- Soluble in water
- **Function:** Enzymatic, transport, hormonal
- **Examples:** Hemoglobin, Enzymes, Insulin

3. BASED ON FUNCTION

STRUCTURAL PROTEINS
Provide strength and support
Examples: Collagen, Keratin

ENZYMATIC PROTEINS
Catalyze biochemical reactions
Examples: Amylase, DNA Polymerase

TRANSPORT PROTEINS
Transport substances across membranes or in blood
Examples: Hemoglobin, Albumin

DEFENSIVE PROTEINS
Protect against diseases
Examples: Antibodies (immuno-globulins)

HORMONAL PROTEINS
Regulate physiological processes
Examples: Insulin, Growth hormone

STORAGE PROTEINS
Store amino acids or ions
Examples: Ferritin (iron), Casein (milk)

CONTRACTILE PROTEINS
Involved in muscle contraction and movement
Examples: Actin, Myosin

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SIMPLE VS CONJUGATED VS DERIVED PROTEINS

SIMPLE PROTEINS	CONJUGATED PROTEINS	DERIVED PROTEINS
Simple proteins consist solely of amino acid chains	Conjugated proteins are proteins that contain both amino acid chains and non-protein components (prosthetic groups)	Derived proteins are molecules that form as a result of the hydrolysis or denaturation of simple or conjugated proteins
Composed only of amino acids	Include additional non-amino acid components such as metal ions, lipids, or carbohydrates	Originate from the breakdown or modification of simpler protein structures
Primarily serve structural roles or act as enzyme	Exhibit diverse functions, often related to the non-protein components, e.g., hemoglobin (iron-containing) for oxygen transport	May retain or lose the original protein's function, depending on the extent of modification

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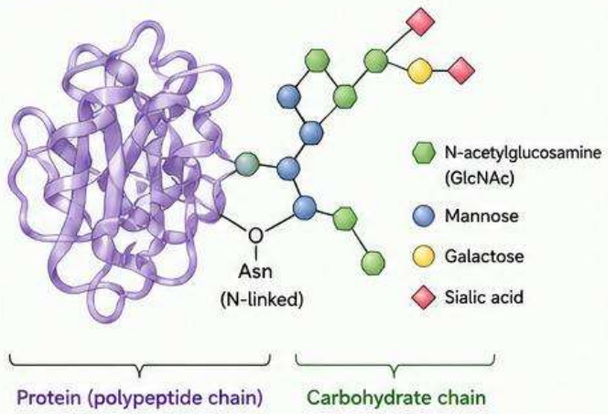
PROTEIN CONJUGATES: STRUCTURE & FUNCTION

1. GLYCOPROTEIN

Protein covalently bonded to one or more **carbohydrate** (oligosaccharide) chains.

Structure



Functions

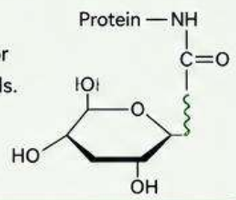
- Cell–cell recognition
- Cell adhesion
- Immune response
- Protein stability and folding

Examples

- Immunoglobulins
- Mucins
- Erythropoietin
- Transferrin

Key linkage

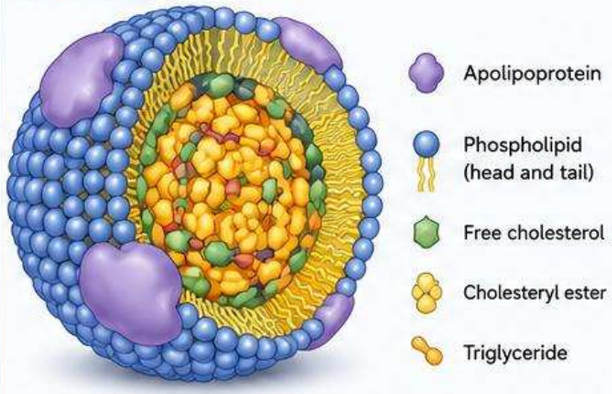
Usually linked via N-linked (Asn) or O-linked (Ser/Thr) glycosidic bonds.



2. LIPOPROTEIN

Protein noncovalently associated with **lipids** to transport **hydrophobic molecules** in aqueous environments.

Structure



Functions

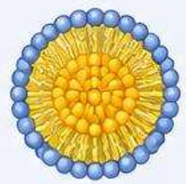
- Transport of lipids (cholesterol, TG, etc.) in blood
- Deliver lipids to tissues
- Maintain lipid homeostasis

Examples

- Chylomicrons
- VLDL, LDL, HDL
- Apolipoprotein B-100
- Apolipoprotein A-I

Key feature

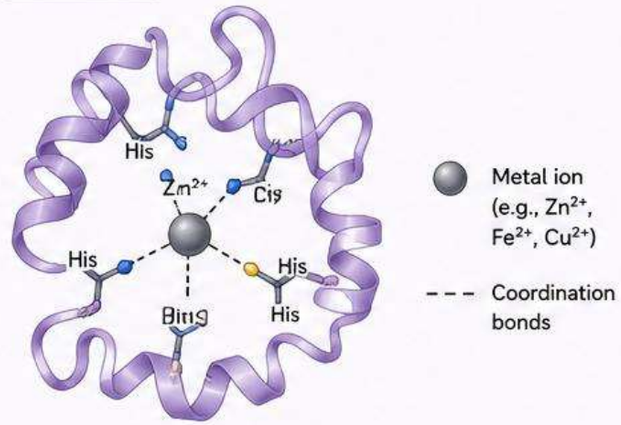
Lipid core is hydrophobic; protein and phospholipid form an amphipathic surface.



3. METALLOPROTEIN

Protein that contains one or more **metal ion(s)** as integral components.

Structure



Functions

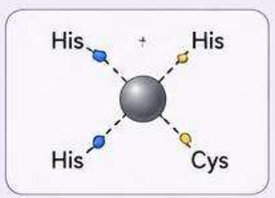
- Catalysis (enzymes)
- Electron transfer
- Oxygen transport
- Structural stability
- Regulation

Examples

- Carbonic anhydrase (Zn^{2+})
- Cytochrome c (Fe)
- Hemoglobin (Fe^{2+})
- Superoxide dismutase (Cu/Zn)
- Alcohol dehydrogenase (Zn^{2+})

Key feature

Metal ion is tightly bound by specific amino acid side chains.



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Protein Classification & Conjugated Proteins

Classification by Composition

- **Simple Proteins:** Composed exclusively of amino acids. Examples include albumins (soluble in water, e.g., serum albumin), globulins (insoluble in water, e.g., antibodies), scleroproteins (structural, e.g., collagen, keratin), and histones (DNA-packaging proteins).
- **Conjugated Proteins:** Contain a protein (apoprotein) and a non-protein prosthetic group. The complete functional molecule is a **holoprotein**. Examples include glycoproteins, lipoproteins, nucleoproteins, metalloproteins, and hemoproteins.

Major Classes of Conjugated Proteins

Glycoproteins

Proteins covalently linked to oligosaccharide chains. Found in cell membranes (receptors, adhesion molecules), blood plasma (immunoglobulins, clotting factors), and extracellular matrix. N-linked glycosylation occurs in the ER; O-linked in the Golgi.

Lipoproteins

Complexes of lipids and apolipoproteins that transport hydrophobic lipids in blood. **HDL** (high-density) removes cholesterol from tissues; **LDL** (low-density) delivers cholesterol to peripheral cells. Imbalance is a major cardiovascular risk factor.

Metalloproteins

Proteins containing metal ion cofactors essential for function. Examples: **hemoglobin** (Fe^{2+} for O_2 transport), **cytochrome c** (Fe for electron transport), **carbonic anhydrase** (Zn^{2+} for CO_2 hydration), and **superoxide dismutase** (Cu/Zn or Mn for antioxidant defense).

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SUMMARY

Key Takeaways: Organic Constituents of Cells

The three major classes of organic cellular constituents — carbohydrates, lipids, and proteins — are interconnected in both structure and function. Together, they govern energy metabolism, membrane architecture, enzymatic catalysis, and cellular communication. A deep understanding of their chemistry is foundational to biochemistry, cell biology, and molecular medicine.

Carbohydrates

Primary energy source (glucose); structural components (cellulose, chitin); cell recognition (glycoproteins, glycolipids). Classified as mono-, di-, oligo-, and polysaccharides. Digestion yields glucose via amylase and disaccharidases.

Lipids

Energy-dense storage (9 kcal/g); membrane bilayer (phospholipids, cholesterol); signaling (steroid hormones, eicosanoids). Defined by hydrophobicity. Fatty acid saturation determines physical properties and health effects.

Proteins

Most functionally diverse biomolecules. Four structural levels (primary → quaternary) determine activity. Enzymes, transporters, receptors, antibodies, and structural proteins. Conjugated proteins (glyco-, lipo-, metallo-) expand functional repertoire.

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Standard Amino Acids

Encoded by the genetic code; 9 are essential and must be obtained from the diet.

9 kcal/g

Lipid Energy Density

More than twice the energy yield of carbohydrates or proteins (4 kcal/g each).

~120g

Daily Brain Glucose

The brain's obligate glucose consumption per day under normal physiological conditions.

20,000+

Human Proteins

Estimated number of distinct proteins encoded by the human genome (proteome).

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